Introduction to Zoology

A Compact Guide to Animal Diversity, Structure, and Ecology

1. What Is Zoology?

Zoology is the branch of biology that studies animals, including their structure, physiology, behaviour, classification, distribution, and evolution. The word comes from the Greek *zoion* (animal) and *logos* (study). The Swedish naturalist Carl Linnaeus formalised binomial nomenclature in the 18th century, giving each species a two-part Latin name such as *Panthera leo* for the lion.

Major sub-disciplines include entomology (insects), ornithology (birds), herpetology (reptiles and amphibians), ichthyology (fishes), mammalogy (mammals), and ethology (animal behaviour). Zoologists work in field research, laboratories, museums, zoos, and conservation agencies.

2. Classification of Animals

All animals belong to the kingdom Animalia. They are multicellular, heterotrophic eukaryotes whose cells lack cell walls, and most can move at some stage of life. Animals are broadly divided into invertebrates, which lack a backbone, and vertebrates, which possess one. Invertebrates make up roughly 95 percent of all known animal species.

Figure 1. Simplified classification of the animal kingdom

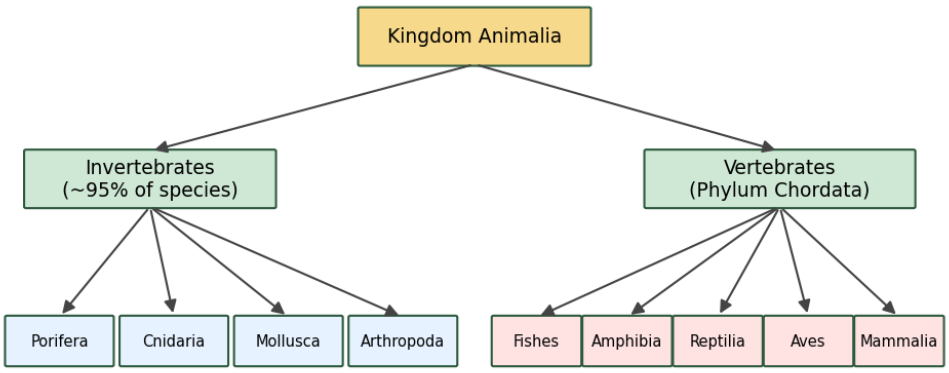


Figure 1. Simplified classification of the animal kingdom

3. Invertebrates

Porifera (sponges) are the simplest animals. They have no true tissues or organs and filter food particles from water drawn through pores in their bodies. **Cnidaria**, including jellyfish, corals, and sea anemones, have radial symmetry and stinging cells called cnidocytes.

Mollusca include snails, clams, and octopuses. Most have a soft body, a muscular foot, and a mantle that often secretes a shell. Cephalopods such as the octopus have the most complex nervous systems among invertebrates and can solve puzzles.

Arthropoda is the largest phylum, with over one million described species. Arthropods have jointed legs, a chitinous exoskeleton, and segmented bodies. The phylum includes insects, arachnids, crustaceans, and myriapods. Insects alone account for more than half of all known living species. Because the exoskeleton cannot grow, arthropods must moult (a process called ecdysis).

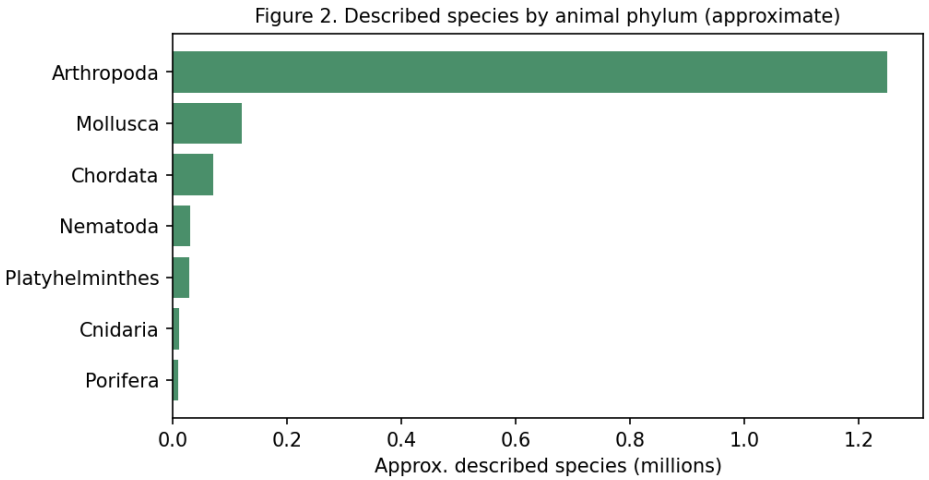


Figure 2. Described species by animal phylum (approximate)

Key facts: Invertebrates

Echinoderms (starfish, sea urchins) have five-part radial symmetry as adults and a water vascular system for locomotion. Annelids (earthworms) have segmented bodies and are important for soil aeration. Nematodes are among the most abundant animals on Earth.

4. Vertebrates and Their Classes

Vertebrates are chordates with a vertebral column that protects the spinal cord. There are five major classes: fishes, amphibians, reptiles, birds, and mammals. Each class shows distinct adaptations in respiration, circulation, and reproduction.

Class	Body temperature	Respiration	Heart chambers	Example
Fishes	Ectothermic	Gills	2	Rohu, shark
Amphibians	Ectothermic	Gills, skin, lungs	3	Frog, salamander
Reptiles	Ectothermic	Lungs	3 (4 in crocodilians)	Cobra, turtle
Birds	Endothermic	Lungs + air sacs	4	Eagle, penguin
Mammals	Endothermic	Lungs	4	Elephant, bat

Table 1. Comparison of vertebrate classes

The evolution of the heart is a classic example of increasing efficiency. In fishes the heart pumps blood in a single circuit through the gills. Amphibians and most reptiles have two atria and one ventricle, allowing partial mixing of oxygenated and deoxygenated blood. Birds and mammals have a fully divided four-chambered heart, which completely separates the two blood streams and supports high metabolic rates. (*Crocodilians are the reptiles with a four-chambered heart.)

Figure 3. Heart chambers across vertebrate classes

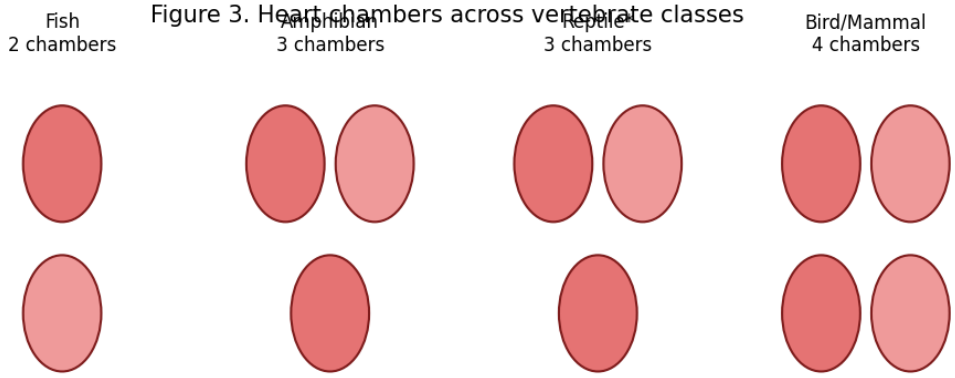


Figure 3. Heart chambers across vertebrate classes

5. Mammals

Mammals are defined by mammary glands that produce milk, hair or fur, three middle-ear bones, and a neocortex in the brain. There are three groups: monotremes (egg-laying, e.g. the platypus and echidna), marsupials (pouched, e.g. the kangaroo), and placentals (e.g. dogs, whales, humans).

The blue whale (*Balaenoptera musculus*) is the largest animal known to have existed, reaching about 30 metres in length and up to roughly 150 tonnes. At the other extreme, the Etruscan shrew weighs only about 2 grams, and the bumblebee bat is among the smallest mammals by skull size.

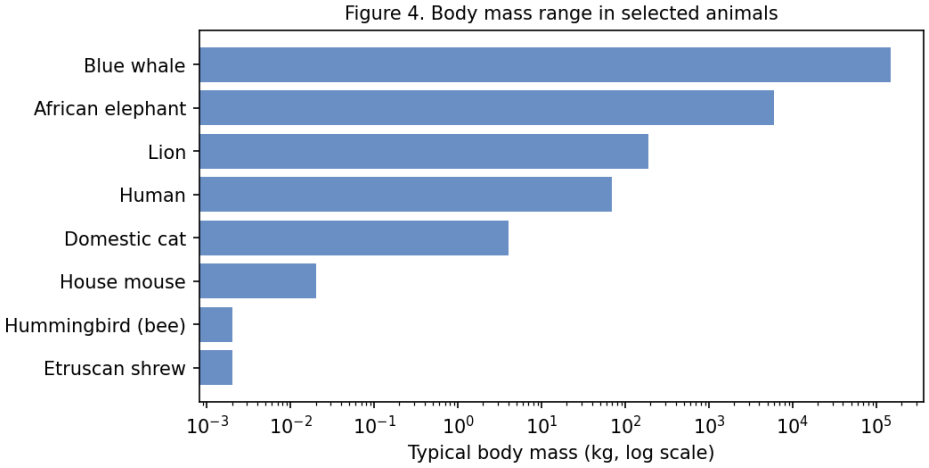


Figure 4. Body mass range in selected animals (log scale)

Birds

Birds (class Aves) have feathers, a beak without teeth, hollow pneumatic bones, and lay hard-shelled eggs. Their unique respiratory system uses air sacs for one-way airflow through the lungs, which is highly efficient at high altitude. The bee hummingbird, found in Cuba, is the smallest bird at about 2 grams, whereas the ostrich is the largest living bird, standing up to 2.7 metres tall.

6. Amphibians and Reptiles

Amphibians such as frogs, toads, newts, and caecilians typically live part of their life in water and part on land. Their moist, permeable skin allows gas exchange, which makes them sensitive indicators of environmental pollution. Most undergo metamorphosis, a dramatic change from a gill-breathing aquatic larva to a lung-breathing adult.

Figure 5. Metamorphosis in the common frog

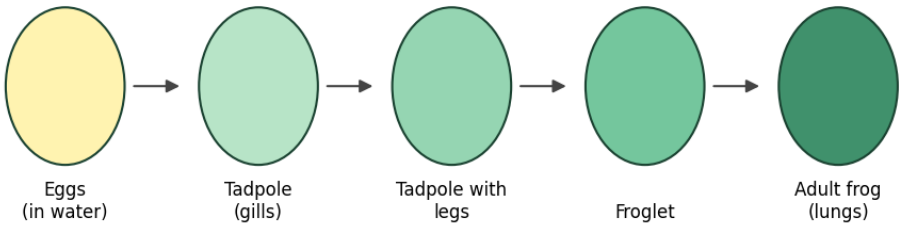


Figure 5. Metamorphosis in the common frog

Reptiles have dry, scaly skin and lay amniotic eggs with leathery shells, which freed them from dependence on water for reproduction. Snakes, lizards, turtles, crocodilians, and the tuatara are living reptiles. As ectotherms, reptiles regulate body temperature behaviourally, for example by basking in the sun. The Komodo dragon (*Varanus komodoensis*) is the heaviest living lizard.

Fishes

Fishes are aquatic vertebrates that breathe through gills. They are divided into jawless fishes (lampreys), cartilaginous fishes (sharks and rays), and bony fishes (the vast majority, such as carp and tuna). Many bony fishes have a swim bladder to control buoyancy, and a lateral line system to sense water movement.

7. Animal Ecology

Ecology examines how animals interact with each other and their environment. Energy enters most ecosystems through producers (plants and algae), which are eaten by primary consumers (herbivores), which in turn are eaten by secondary and tertiary consumers (carnivores). Decomposers such as fungi and bacteria recycle nutrients. Only about 10 percent of energy is transferred from one trophic level to the next, which is why food chains are usually short.

Figure 6. A simple food web (arrows show energy flow)

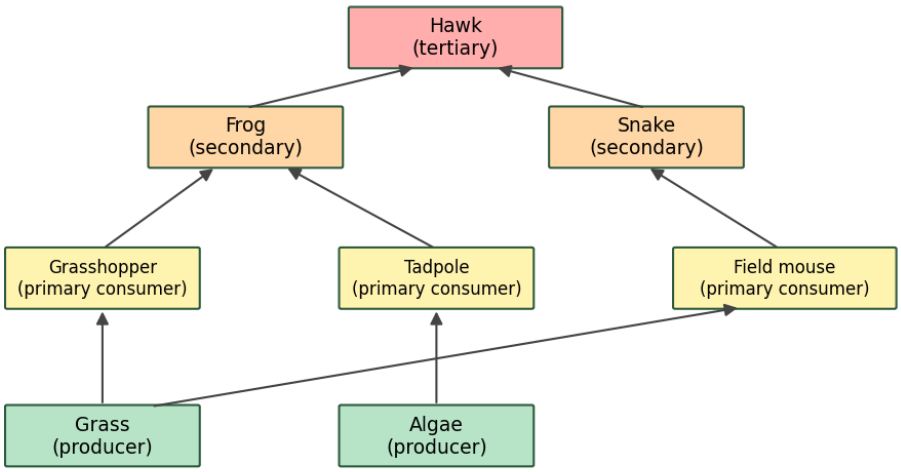


Figure 6. A simple food web (arrows show energy flow)

Important ecological relationships include **predation**, **competition**, **commensalism** (one benefits, the other is unaffected), **mutualism** (both benefit, e.g. clownfish and sea anemones), and **parasitism** (one benefits at the other's expense, e.g. tapeworms in the gut of a host).

Adaptations

Adaptations are heritable traits that improve survival and reproduction. Examples include camouflage in stick insects, migration in Arctic terns (which travel roughly 70,000 km each year between the Arctic and Antarctic), hibernation in bears, and echolocation in bats and dolphins.

8. Conservation

Habitat loss, overexploitation, pollution, invasive species, and climate change are the main threats to animal biodiversity. The International Union for Conservation of Nature (IUCN) maintains the Red List, which classifies species by extinction risk. The Bengal tiger, the giant panda, and the Asian elephant are well-known examples of species that have been the focus of conservation programmes. In India, Project Tiger was launched in 1973 to protect the tiger and its habitat.

Figure 7. IUCN Red List categories (low to high extinction risk)

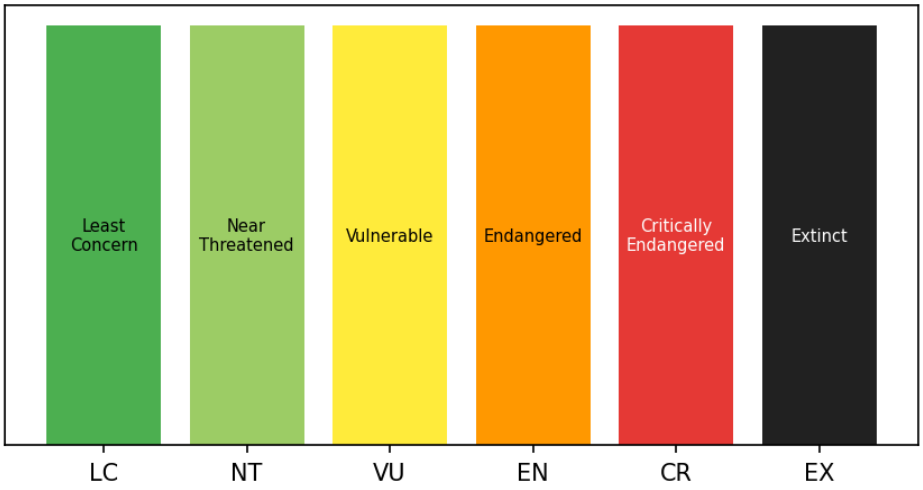


Figure 7. IUCN Red List categories

Quick Review Questions (for RAG testing)

1. What is the largest phylum of animals? *Arthropoda.*
2. How many heart chambers do birds have? *Four.*
3. What is ecdysis? *The moulting of the exoskeleton in arthropods.*
4. Which animal is the largest known to have existed? *The blue whale.*
5. What year was Project Tiger launched in India? *1973.*
6. What are the three groups of mammals? *Monotremes, marsupials, and placentals.*
7. What percentage of energy passes between trophic levels? *About 10 percent.*

Glossary

Ectotherm: animal that relies on external heat sources. **Endotherm:** animal that generates its own body heat. **Metamorphosis:** major developmental change in body form. **Trophic level:** position in a food chain. **Taxonomy:** science of naming and classifying organisms.